

Aleksei Liuliakov

PhD in Machine Learning, dissertation submitted September 2026, defence expected December 2026

- **DGPO** (IJCNN 2026): RL-steered discrete **graph diffusion** for **neural architecture generation**; matches the NAS-Bench-201 optima on all three tasks (CIFAR-10 91.61%, CIFAR-100 73.49%, ImageNet-16-120 46.77%; by benchmark lookup); pretrained on 7% of NAS-Bench-101, the best generated architecture lands within 0.32 pp of the full-data model. [\[arXiv\]](#) [\[code\]](#)
- **Anomaly detection on dynamic graphs**: TGN-SVDD reaches ROC-AUC 0.994–0.999 vs 0.268–0.690 for vanilla TGN on CIC-IDS2017 (ICANN 2023); probabilistic RTGN-SVDD raises ROC-AUC 0.574 → 0.844 and 0.561 → 0.880 under 50% label contamination vs the deterministic model (ESANN 2024).
- **Multi-objective AutoML** (TPOT, Optuna, NSGA-II): Pareto fronts between accuracy and sparsity show that different architectures are optimal at different sparsity levels for intrusion detection (UNSW-NB15, NSL-KDD) and outlier detection (Applied Soft Computing, 2023).

Experience

- Nov 2020 – Present **Doctoral Researcher – Machine Learning, Bielefeld University, Bielefeld, Germany**
- DGPO: extended RL-steered discrete graph diffusion (DiGress + REINFORCE-style policy optimization) from undirected molecular graphs to DAGs via topological node ordering and positional encoding; 94.50% on NAS-Bench-101 (benchmark lookup); multi-objective MO-DGPO reaches Pareto hypervolume 0.9901 on NAS-Bench-201, matching the union of three single-task runs.
 - Failure-mode check: bidirectional-control ablation, where inverse reward drives accuracy to 9.5% (chance), validating directional control of the generator.
 - Ran multi-seed training and evaluation campaigns on SLURM GPU clusters (3 seeds; single-GPU, ~115 GPU-hours for DGPO on one A40).
 - Engineering: public research repos [DGPO](#) (MIT, Makefile one-command reproduction, pinned requirements) and [rtgn_svdd](#) (pyproject, pytest tests, docs).
 - Published 6 papers (5 as first author); built AI-augmented research workflows (Claude Code, Codex) to automate experimental pipelines, from implementation through systematic evaluation.
- Oct 2023 – Sep 2025 **Tutor & Supervisor, Bielefeld University**
- Applied Optimization tutorials (coordinated fellow tutors); sole seminar instructor, 'Implications of the AI Act in Machine Learning'.
- Sep 2011 – Jul 2018 **Founder & CEO – Baltik Bread, Bakery Manufacturing & Mobile Retail, Novosibirsk, Russia**
- Co-founded and scaled a dual manufacturing + retail operation from zero; 500k-600k EUR investment; full ROI within 4 years.
 - Grew to ~1M EUR annual revenue, ~2,500 units/day across ~40 SKUs with 90%+ sell-through; built and managed a 25-person team.
 - Data-driven location optimization using empirical sales analysis; built competitive advantage through systematic testing.

Education

- Nov 2020 – Sep 2026 **PhD (Dr. rer. nat.) in Machine Learning, Bielefeld University, Bielefeld, Germany**
Dissertation submitted Sep 2026, defence expected Dec 2026. Supervisor: Prof. Barbara Hammer.
- Oct 2018 – Mar 2021 **M.Sc. Data Science, Bielefeld University, Bielefeld, Germany**
- Sep 2007 – Jun 2010 **M.Sc. Econometrics & Quantitative Economics, Novosibirsk State University, Novosibirsk, Russia**
- Sep 2003 – Jun 2007 **B.Sc. Physics, Novosibirsk State University, Novosibirsk, Russia**

Publications

- **A. Liuliakov**, L. Hermes, and B. Hammer. “**DGPO: RL-Steered Graph Diffusion for Neural Architecture Generation.**” In: *IJCNN 2026 (WCCI)*. [arXiv] [code]
- **A. Liuliakov**, A. Schulz, L. Hermes, and B. Hammer. “**Noise Robust One-Class Intrusion Detection on Dynamic Graphs.**” In: *ESANN 2024*. [arXiv] [code]
- **A. Liuliakov**, A. Schulz, L. Hermes, and B. Hammer. “**One-Class Intrusion Detection with Dynamic Graphs.**” In: *ICANN 2023*. [arXiv] [code]
- L. Hermes, **A. Liuliakov**, and M. Schilling. “**Graph Learning by Dynamic Sampling.**” In: *IJCNN 2023*.
- **A. Liuliakov**, L. Hermes, and B. Hammer. “**AutoML Technologies for the Identification of Sparse Classification and Outlier Detection Models.**” In: *Applied Soft Computing*, 2023.
- **A. Liuliakov** and B. Hammer. “**AutoML Technologies for the Identification of Sparse Models.**” In: *IDEAL 2021*.

Service Session chair, IJCNN 2026 (IEEE WCCI); peer reviewing: ESANN 2024 (1), IJCNN 2024 (1), ESANN 2026 (2), IJCNN 2026 (2) papers.

Skills

ML & AI	PyTorch, Diffusion Models, Reinforcement Learning, Graph Neural Networks, AutoML / Neural Architecture Search, Generative AI, Deep Learning, Anomaly Detection, OOD Robustness, Probabilistic Modeling
Stack	Python, PyTorch Geometric, PyTorch Lightning, Optuna, scikit-learn, NumPy, Pandas; Linux clusters, Slurm, Docker, W&B, Git/GitHub
AI Tools	Claude Code, OpenAI Codex, GitHub Copilot; agentic workflows for experiment automation and GPU cluster deployment via SLURM

Languages

English (C1), German (B1), Russian (native), French (A1)